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1 /Users/mehakagrawal/PycharmProjects/
  Final_Dissertation/.venv/bin/python /Users/
  mehakagrawal/Desktop/Final_Dissertation/Codes/2.
  q2_age_outlier_sensitivity/q2_outlier_sensitivity.
  py
2 Gene universe after filtering: 22253 genes (shared
  by both runs)
3
4
5 DESeq2 - Q2 (FULL / with outliers): 11 samples
6   old (M37_WT)=6   young (M11_WT)=5
7 Fitting size factors...
8 ... done in 0.01 seconds.
9
10 Using None as control genes, passed at DeseqDataSet
   initialization
11 Fitting dispersions...
12 ... done in 3.12 seconds.
13
14 Fitting dispersion trend curve...
15 ... done in 0.20 seconds.
16
17 Fitting MAP dispersions...
18 ... done in 2.10 seconds.
19
20 Fitting LFCs...
21 ... done in 1.69 seconds.
22
23 Calculating cook's distance...
24 ... done in 0.01 seconds.
25
26 Replacing 0 outlier genes.
27
28 Running Wald tests...
29 ... done in 0.52 seconds.
30
31 Log2 fold change & Wald test p-value: age_group old
   vs young
32           baseMean  log2FoldChange
   ...      pvalue      padj
33 gene_id

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33
...
34 ENSDARG00000040565 2.000299e+06 -0.220466
... 0.782353 0.999934
35 ENSDARG00000063905 1.340261e+06 -1.384314
... 0.088427 0.748012
36 ENSDARG00000035438 1.054547e+06 -0.460944
... 0.505152 0.999934
37 ENSDARG00000099420 7.854702e+05 -0.099829
... 0.896639 0.999934
38 ENSDARG00000024433 7.862142e+05 -0.247169
... 0.758905 0.999934
39 ...
...
40 ENSDARG00000038191 4.658030e-01 0.947596
... 0.751938 0.999934
41 ENSDARG00000097802 1.838813e+00 0.925437
... 0.656391 0.999934
42 ENSDARG00000056652 0.000000e+00 NaN
... NaN NaN
43 ENSDARG00000043584 1.150212e+00 3.277445
... 0.319317 0.999934
44 ENSDARG00000096866 2.421524e+00 3.030638
... 0.172975 0.968128
45
46 [22253 rows x 6 columns]
47 DEGs (q<0.05, |log2FC|>1.0): 515 (up in old=368
, down in old=147) [pi0=1.000]
48
49
50 DESeq2 - Q2 (REDUCED / outliers dropped): 8 samples
51 old (M37_WT)=4 young (M11_WT)=4
52 Using None as control genes, passed at DeseqDataSet
initialization
53 Fitting size factors...
54 ... done in 0.00 seconds.
55
56 Fitting dispersions...
57 ... done in 1.71 seconds.
58

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59 Fitting dispersion trend curve...
60 ... done in 0.27 seconds.
61
62 Fitting MAP dispersions...
63 ... done in 1.94 seconds.
64
65 Fitting LFCs...
66 ... done in 1.45 seconds.
67
68 Calculating cook's distance...
69 ... done in 0.01 seconds.
70
71 Replacing 0 outlier genes.
72
73 Running Wald tests...
74 ... done in 0.55 seconds.
75
76 Log2 fold change & Wald test p-value: age_group old
    vs young
77
78
79
80
81
82
83
84
85
86
87

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gene_id	pvalue	padj	baseMean	log2FoldChange
ENSDARG00000040565	0.953128	0.990709	2.171070e+06	0.020915
ENSDARG00000063905	0.000051	0.001853	1.487143e+06	-1.109652
ENSDARG00000035438	0.399848	0.765270	1.137054e+06	-0.236707
ENSDARG00000099420	0.473336	0.811816	8.599801e+05	0.169870
ENSDARG00000024433	0.914836	0.980660	8.683547e+05	0.025000
ENSDARG00000038191	0.770513	NaN	4.784839e-01	1.044537
ENSDARG00000097802	0.646876	NaN	2.032951e+00	1.122607
ENSDARG00000056652	0.000000e+00		0.000000e+00	NaN

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87      ...      NaN      NaN
88 ENSDARG00000043584  1.321998e+00      3.722120
      ...  0.309603      NaN
89 ENSDARG00000096866  1.453244e+00      2.156422
      ...  0.527659      NaN
90
91 [22253 rows x 6 columns]
92 DEGs (q<0.05, |log2FC|>1.0): 1219 (up in old=
    785, down in old=434) [pi0=0.834]
93
94
95 DEG SET COMPARISON
96 Full DEGs      : 515
97 Reduced DEGs   : 1219
98 Shared         : 459
99 Full-only (lost) : 56
100 Reduced-only(new): 760
101 Jaccard        : 0.360
102 Recovery of full set in reduced run: 89.1%
103
104
105 log2FC CONCORDANCE (genes significant in EITHER
    run)
106 Pearson r      : 0.960
107 Spearman r     : 0.970
108 Same sign      : 97.8%
109
110
111 TOP-20 FULL-RUN HITS: still significant after
    dropping outliers?
112 [kept ] si_dkey-250k10.1      log2FC +9.08 -> +9.
    54 q_red=1.30e-50
113 [kept ] znf1179              log2FC -5.88 -> -5.
    66 q_red=2.11e-46
114 [kept ] BX927329.1          log2FC -8.47 -> -8.
    93 q_red=8.33e-09
115 [kept ] mhc1uba              log2FC -11.97 -> -
    11.59 q_red=1.79e-10
116 [kept ] si_ch211-214b16.4    log2FC -7.39 -> -7.
    15 q_red=1.99e-34
117 [kept ] nub1                log2FC -4.28 -> -4.

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117 26 q_red=1.54e-10
118 [kept ] apoa1b log2FC +11.90 -> +
12.32 q_red=6.61e-14
119 [kept ] si_dkey-10p5.10 log2FC -4.25 -> -5.
07 q_red=5.51e-24
120 [kept ] si_dkey-238k10.1 log2FC +7.42 -> +8.
06 q_red=5.33e-21
121 [kept ] fetub log2FC +11.19 -> +
12.05 q_red=3.53e-12
122 [kept ] FP074902.2 log2FC -5.47 -> -4.
93 q_red=2.22e-16
123 [kept ] fga log2FC +11.07 -> +
11.85 q_red=2.01e-10
124 [kept ] apobb.1 log2FC +9.77 -> +10
.14 q_red=1.78e-12
125 [kept ] arl4d log2FC +3.43 -> +3.
48 q_red=1.86e-08
126 [kept ] thumpd3 log2FC -9.09 -> -8.
42 q_red=2.13e-07
127 [kept ] pvalb7 log2FC -5.18 -> -4.
93 q_red=2.41e-56
128 [kept ] c3a.2 log2FC +10.42 -> +
11.59 q_red=1.75e-07
129 [kept ] tfa log2FC +9.29 -> +10
.94 q_red=2.77e-13
130 [kept ] mospd1 log2FC -3.72 -> -3.
53 q_red=3.04e-07
131 [kept ] si_dkey-71b5.7 log2FC -4.15 -> -3.
90 q_red=1.65e-21
132
133 20/20 of the top hits remain significant in the
reduced run.
134
135 Saved figure: /Users/mehakagrawal/Desktop/
Final_Dissertation/Outputs/2.
q2_age_outlier_sensitivity/q2_outlier_sensitivity.
png
136 Saved: q2_outlier_sensitivity_comparison.csv
137 Saved: q2_age_DEG_results_no_outliers.csv
138
139 Sensitivity analysis complete.

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140

141 Process finished with exit code 0

142